INTERACTIVE DEMO · PAPER · OPEN CODE

AI breaks rules with total confidence.

I built a way to **catch it — and fix it** — using an idea borrowed from physics.

Swipe →

Teo Qing Cong Eugene

THE PROBLEM

Trained to sound right, not to be right.

Neural networks optimise for what **looks** plausible. Nothing inside them forbids a **logically impossible** answer — an invalid configuration, a self-contradiction.

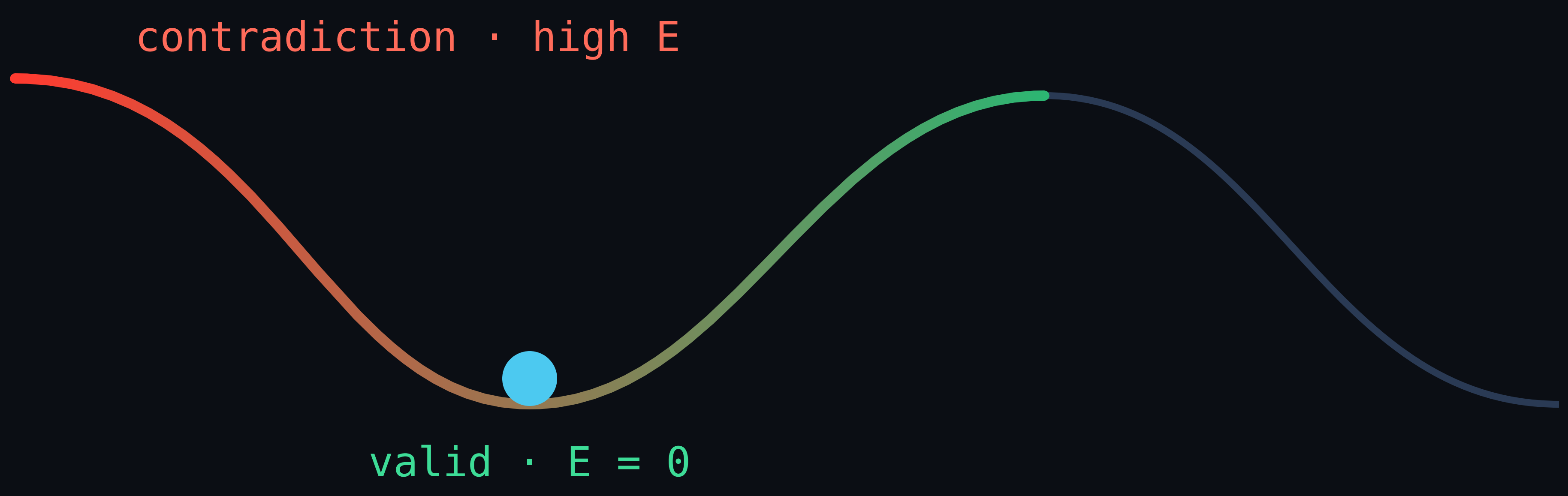
Worse: they hand it to you with **full confidence** and **no signal** that anything is wrong.

THE IDEA

Reasoning as thermodynamics

Give every answer an **energy**. Obey the rules → **0 energy** (a valley). Break one → **high energy** (a hill). "Thinking" is letting the answer roll **downhill** to a valley.

(An analogy for intuition — not literal physics.)



HOW IT WORKS

Score → Explain → Repair

1 · SCORE

Measure how much an answer **breaks your rules**. Zero means it obeys all of them — no labels required.

2 · EXPLAIN

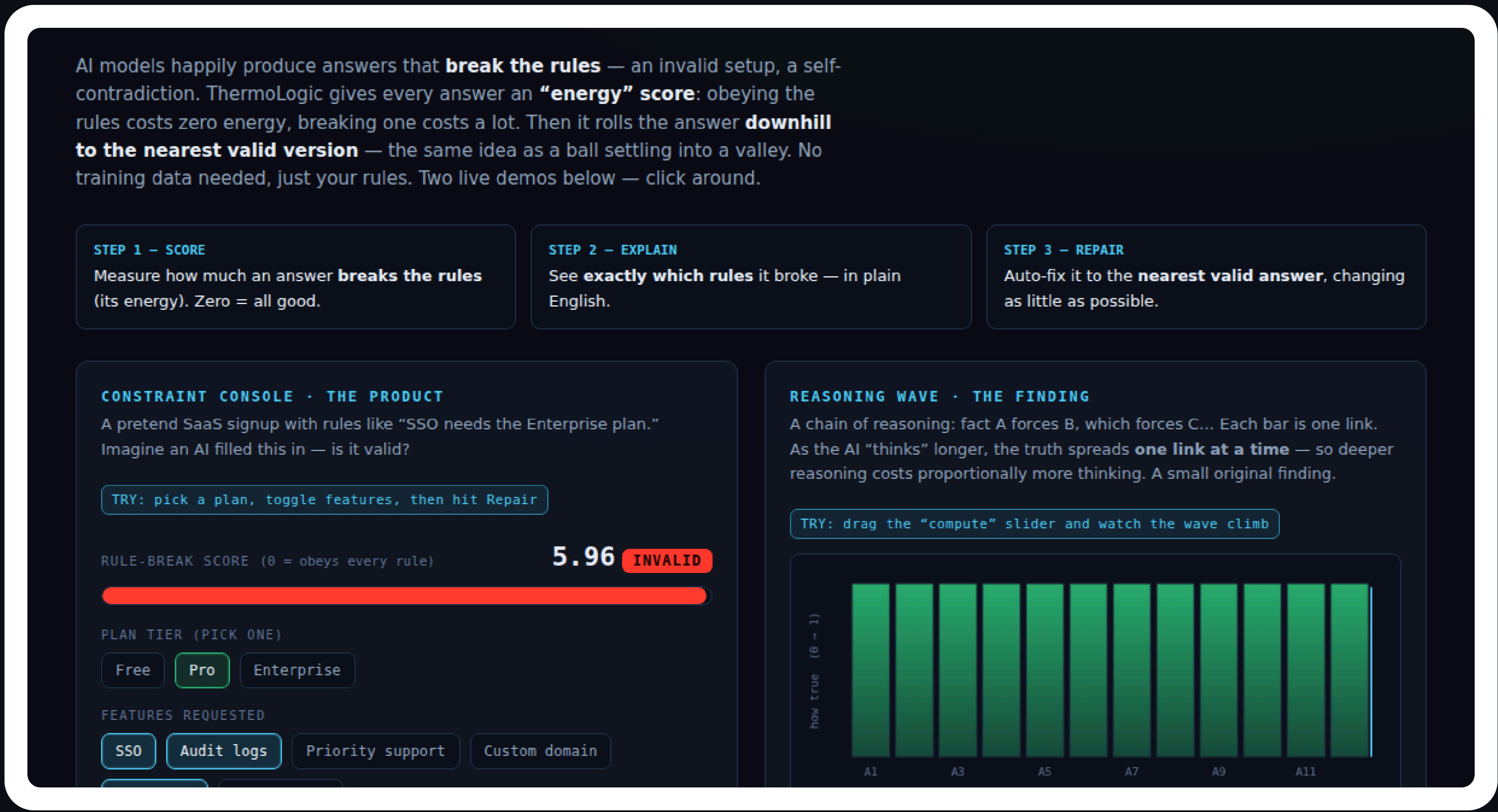
Surface **exactly which rules** it broke, in plain English.

3 · REPAIR

Roll it downhill to the **nearest valid answer**, changing as little as possible.

LIVE IN THE BROWSER

A working demonstrator



Toggle a configuration; the gauge scores validity live; one click repairs it to a valid state — the same engine as the PyTorch library, running client-side.

EXAMPLE · AUTO-REPAIR

An invalid signup, fixed

✗ BEFORE — BREAKS A RULE

Pro plan · requests SSO, Audit logs, Seats>5

→ SSO needs the Enterprise plan

▼ repair (energy descent)

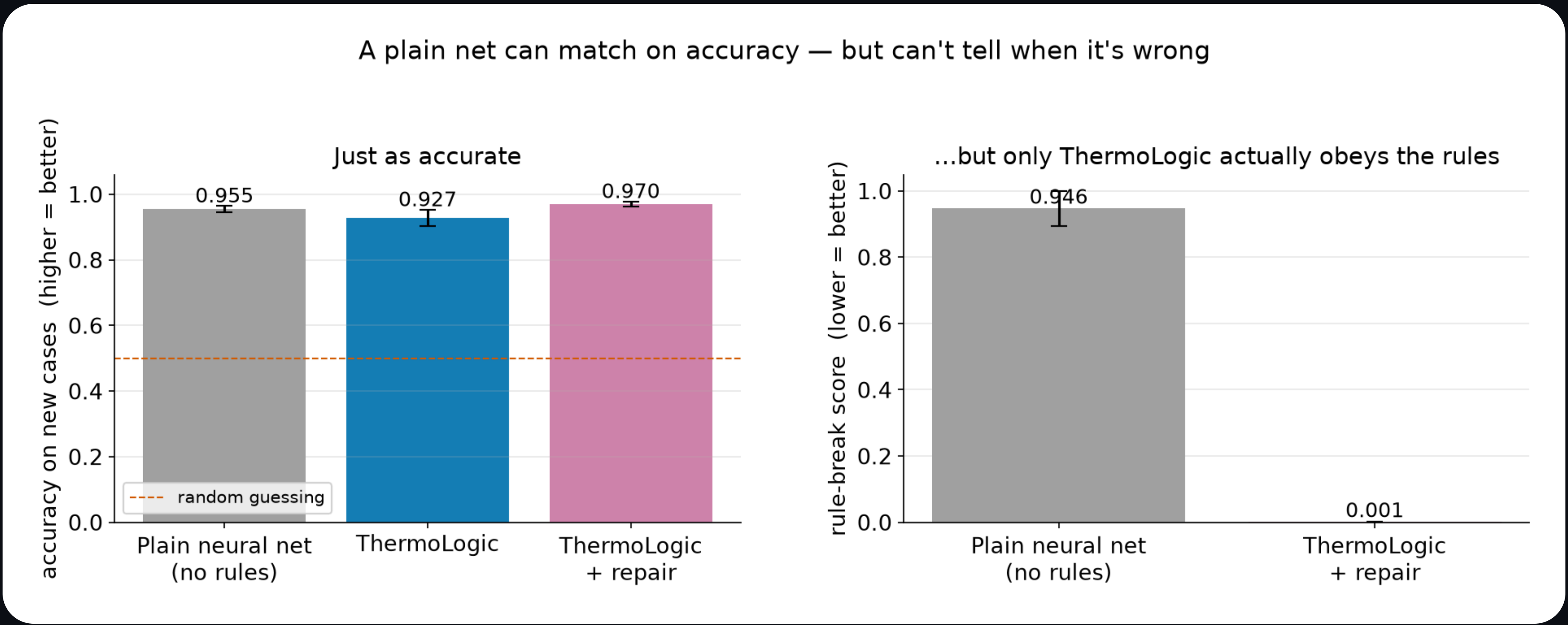
✓ AFTER — VALID

Pro plan · Audit logs, Seats>5 · SSO removed

→ obeys every rule, minimal change

FINDING 1

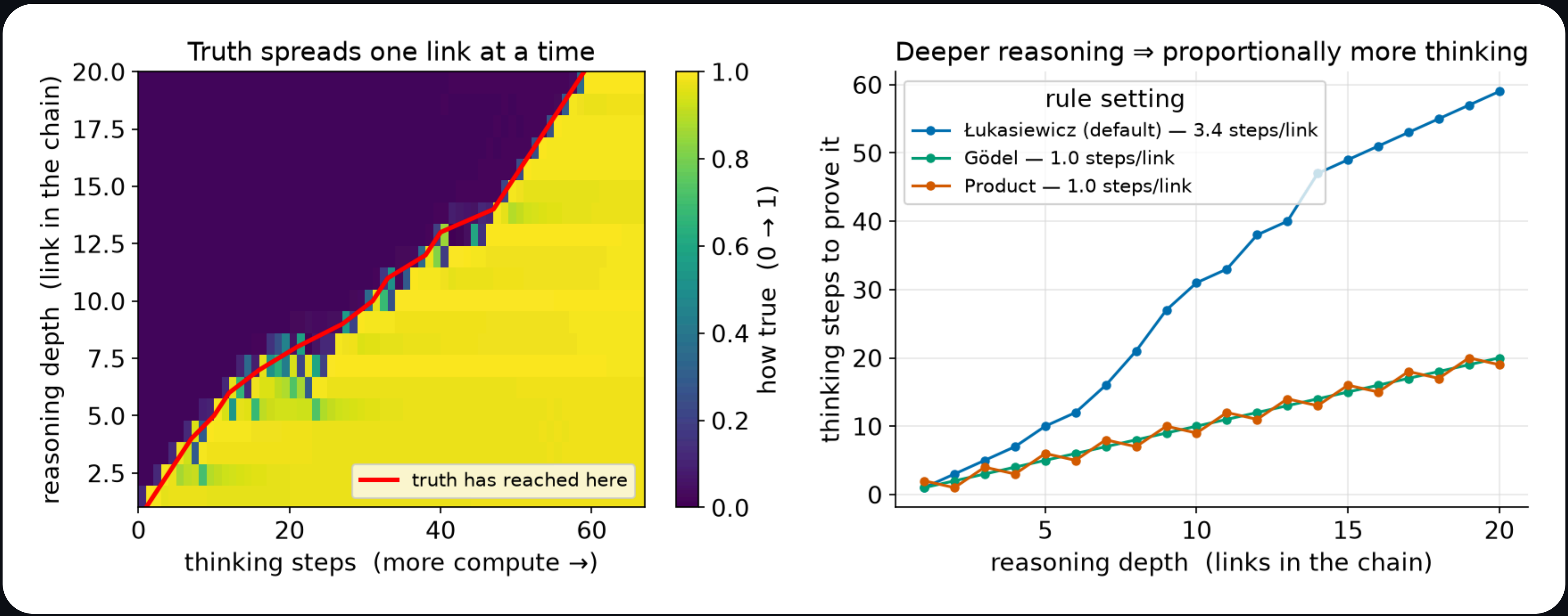
The value is **reliability**, not accuracy



A plain neural net matches on accuracy (0.955) — but its answers are logically **inconsistent**. Only the energy layer gives a label-free **consistency guarantee** (0.001).

FINDING 2

Deeper reasoning = more compute



Watching the energy descent, truth propagates through a chain of rules **one hop at a time** — so twice the reasoning depth costs about twice the "thinking." A small original finding.

THE HONEST SCORECARD

Where it helps — and where it doesn't

✅ **Guarantees consistency** — flags & repairs rule-breaks a plain net can't even detect

✅ **Scales linearly** where an exact solver explodes (2^d), and is **differentiable** — trainable through

✅ **+26 pts** as a semi-supervised signal on a hard rule, when labels are scarce

⚖️ **Ties** — does *not* beat an exact solver on raw accuracy

❌ **No free lunch** — with full labels and easy rules, it's a wash

STRAIGHT TALK

What this is (and isn't)

IT IS

A reproducible **proof-of-concept** and a small usable tool. Builds openly on prior neuro-symbolic work (LTN, DeepProbLog, Semantic Loss). Every claim tied to a number.

IT ISN'T

An LLM upgrade, or a claim to have "solved" hallucination. It works today on **structured outputs with clear rules** — configs, forms, tabular decisions.

Scaling the idea toward language models is the road ahead — the vision, not a result.

TRY IT · READ IT · BUILD WITH IT

Have a look 👇

🕹️ Live demo joperjoker.github.io/Project-ThermoLogic

📄 Technical report (PDF) in the repo

💻 Code github.com/joperjoker/Project-ThermoLogic

📦 pip install -e . from thermologic import LogicEnergy

If your team fights constraint-satisfaction on AI outputs — configs, forms, compliance — I'd love to compare notes.

— Teo Qing Cong Eugene · linkedin.com/in/eugene-teo